



Tho DANG

Robotics/Automatic/AI (PhD.)

► French/Vietnamese

Skills

Linux, Windows, Office 10+ yrs.

Matlab/Simulink 10+ yrs.

Python/C++ 8+ yrs.

ROS/Gazebo/MuJoCo 4+ yrs.

Git/GitLab/Docker 4+ yrs.

CAD 8+ yrs.

Languages

English

French

Vietnamese

Education

06/2021 - present

PhD in Robotics

Montpellier University, France

Robotics

Biography

I obtained Engineering degree in electrical-electronic-control, and Master degree in control in 2003 and in 2006 in Vietnam, respectively. I got PhD degree of the Montpellier University in robotics in June, 2021. My interests are design, control, optimization, and artificial intelligence for robots, robotics software. More information can be found on the website: <https://thodangum.github.io/>.

Work experiences

Researcher

05/2026 - present

Dexter Team

LIRMM, Montpellier University, France

<http://www.lirmm.fr/>

INTELLISHARE project: *Propose a shared control approach for human-robot interaction (Game Theory and Predictive control).*

Key Techniques: Shared control; Game Theory; Model Predictive Control.

Tools: MuJoCo, ROS2, C++/Python.

Researcher

09/2024 - 31/08/2025

Explore Team

Montpellier University, France

LIRMM, Montpellier University, France

<http://www.lirmm.fr/>

Artificial Intelligence for underwater robots:

+ *Reinforcement Learning methods for robots: A Survey.*

+ *Realize localization algorithm for indoor robots.*

+ *Propose a neural network to estimate inverse dynamics model and control an underwater robot, BlueROV.*

Key Techniques: Robot simulation, Artificial intelligence, Reinforcement learning.

Tools: Matlab/C++/Python; ROS2/Gazebo; TensorFlow/PyTorch.

Lecturer - Researcher

09/2022 - 31/08/2024

Teaching: Polytech Montpellier

<https://www.polytech.umontpellier.fr/>

Research: Explore Team - Underwater Robots

LIRMM, Montpellier University, France

<http://www.lirmm.fr/>

Research: Reinforcement Learning, Artificial Intelligence, Neural Network, Regression Algorithms, Inverse Problem:

+ *Propose Actor - Critic architecture to control underwater robots.*

+ *Propose a neural network architecture to estimate inverse model and control an underwater robot.*

Teaching: Mobile Robot, ROS project, Manipulation Robot, Robotic Projects, Linear Multivariable Systems, Discrete Systems, Network and Automatic.

Tools: Matlab/C++/Python; ROS; CoppeliaSim; TensorFlow/PyTorch.

Researcher

01/2022 - 31/08/2022

Explore Team (Robot for Karst exploration)

LIRMM, Montpellier University, France

<http://www.lirmm.fr/>

LEZ 2020 project for Karst exploration: HIL simulation, Underwater robot control - Télémaque : *realize the controller of Télémaque (by Quaternions) which is used for Karst exploration.*

Key Techniques: Quaternion control, Backstepping; Kalman filter.

Tools: C++/Python.

09/2017 - 05/2021

PhD candidate

Montpellier University, France

Robotics

2004 - 2006

Master of Science

Vietnam National University, Vietnam

Control.

1998 - 2003

Bachelor of Engineering

Vietnam National University, Vietnam

Electrical, Electronic and Control.

Contacts

☎ 07.58.55.48.97

✉ danghuu@lirmm.fr

✉ dhtho10@gmail.com

Researcher

Explore Team (Surface Autonomous Vehicles)
LIRMM, Montpellier University, France
<http://www.lirmm.fr/>

06/2021 - 31/12/2021

Energy - Efficient Path Planning for Surface Autonomous Vehicles:
+ *Propose a simulator to find energy-efficient path with respect to energy (consumption et production) a surface autonomous vehicle equipped by a kite, of solar panels and a propulsion system.*
+ *Implement an A* based algorithm to find a best path that optimizes consumption and production energies.*

Key Techniques: Path planning, A* algorithm, Energy model.

Tools: Matlab.

Ph.D Candidate

Explore Team (Prof. Lionel Lapierre - Thesis Advisor)
LIRMM, Montpellier University, France
<http://www.lirmm.fr/>

2017 - 05/2021

Underwater Robots for Karst and Marine Explorations: Study of redundant systems:

+ *Propose performance indices to design optimal configurations of an underwater robot.*

+ *Optimize the configuration of a robot (static et dynamic) in function of performance indices.*

+ *Concept and design the umbrella robot with dynamic configuration (from hardware to software) which can vary its configuration with respect to missions.*

+ *Propose a real-time control allocation approach to optimize energy consumption of the robot. Kalman filter, IMU calibration, and 3D path following method have been implemented.*

+ *Design a controller by quaternions.*

Key Techniques: Optimization; Multiobjective optimization; Path following; Kalman filter; PID, Quaternion control; Backstepping; Control allocation.

Tools: Matlab/C++/Python; FreeCAD/AutoCAD; ANSYS; Electronics.

Researcher

Electrical and Electronic Department
Universiti Teknologi PETRONAS, Malaysia
<https://www.utp.edu.my>

2015 - 2017

Research: Nonlinear System and Model Predictive Control:

+ *Model the system "Twin-Rotor".*

+ *Propose a model predictive controller to do trajectory tracking.*

Teaching: PID regulation, Random Process and Probability.

Key Techniques: System Identification; Model Predictive Control; Optimization.

Tools: Matlab/Simulink/C++/Python.

Lecturer

Electrical and Electronic Department
Ho Chi Minh University of Transport, Vietnam
<https://ut.edu.vn/en/>

2007 - 2015

Teaching: Robotic; Control and measurement systems, system identification, industrial networks, SLAM; linear and nonlinear systems, sensors and motors, PLC (Programmable Logical Controller).

Key Techniques: SLAM; Robotics, Control; Sensors; Motors; PLC.

Tools: Matlab/Simulink/C++/Python; Step7 - Siemens and Schneider PLCs

Engineer

Southern Airport Authority, Vietnam

2003 - 2007

PLC (Programmable Logical Controller), Supervision system, Electrical distribution system: *PLC Program*.

Key Techniques: Robotics; Energy system.

Tools: PLCs of Siemens and Schneider.